nRF52840 Bluetooth Sensor Hub

Headless IoT Platform with NFC-Assisted Secure Pairing

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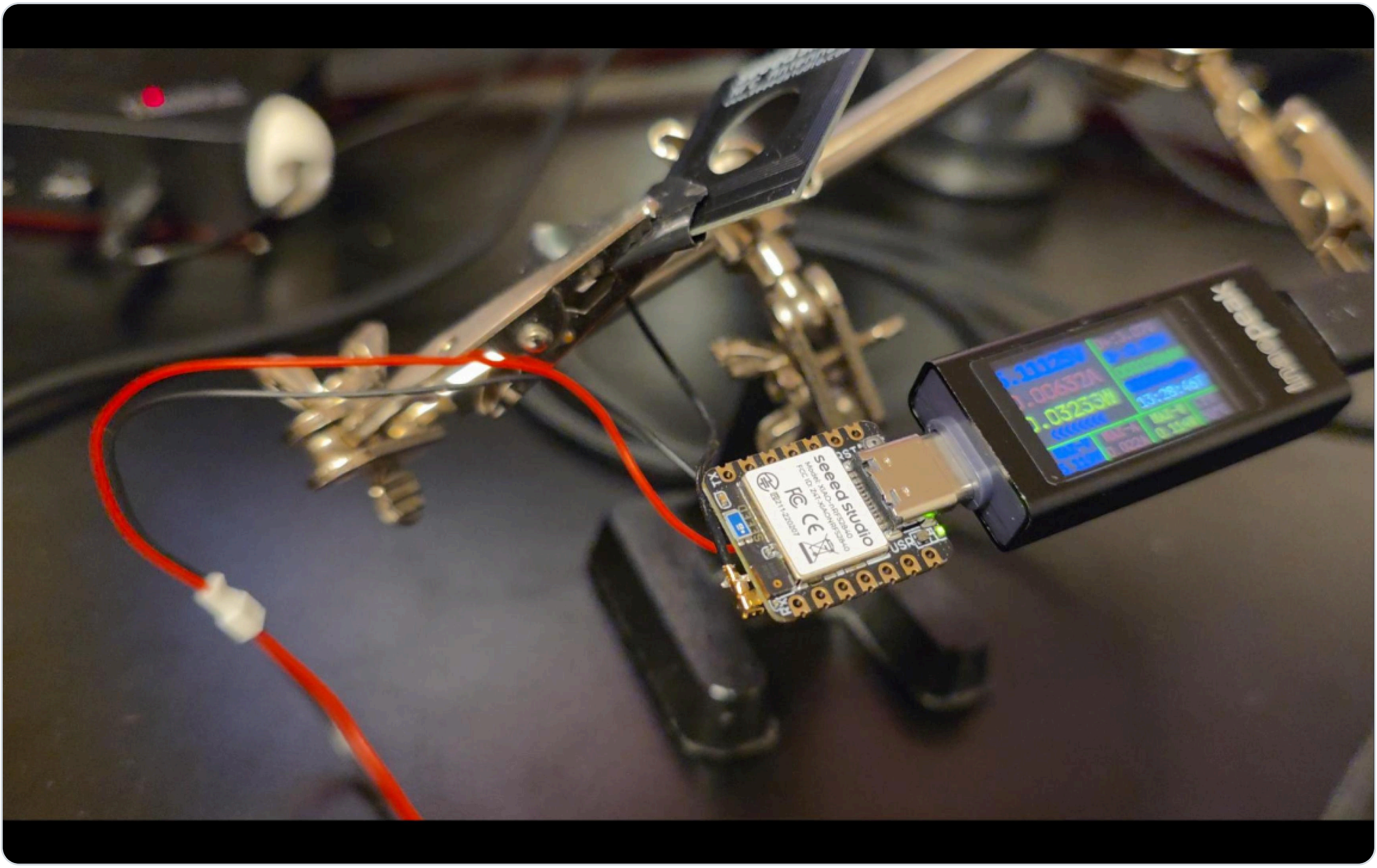
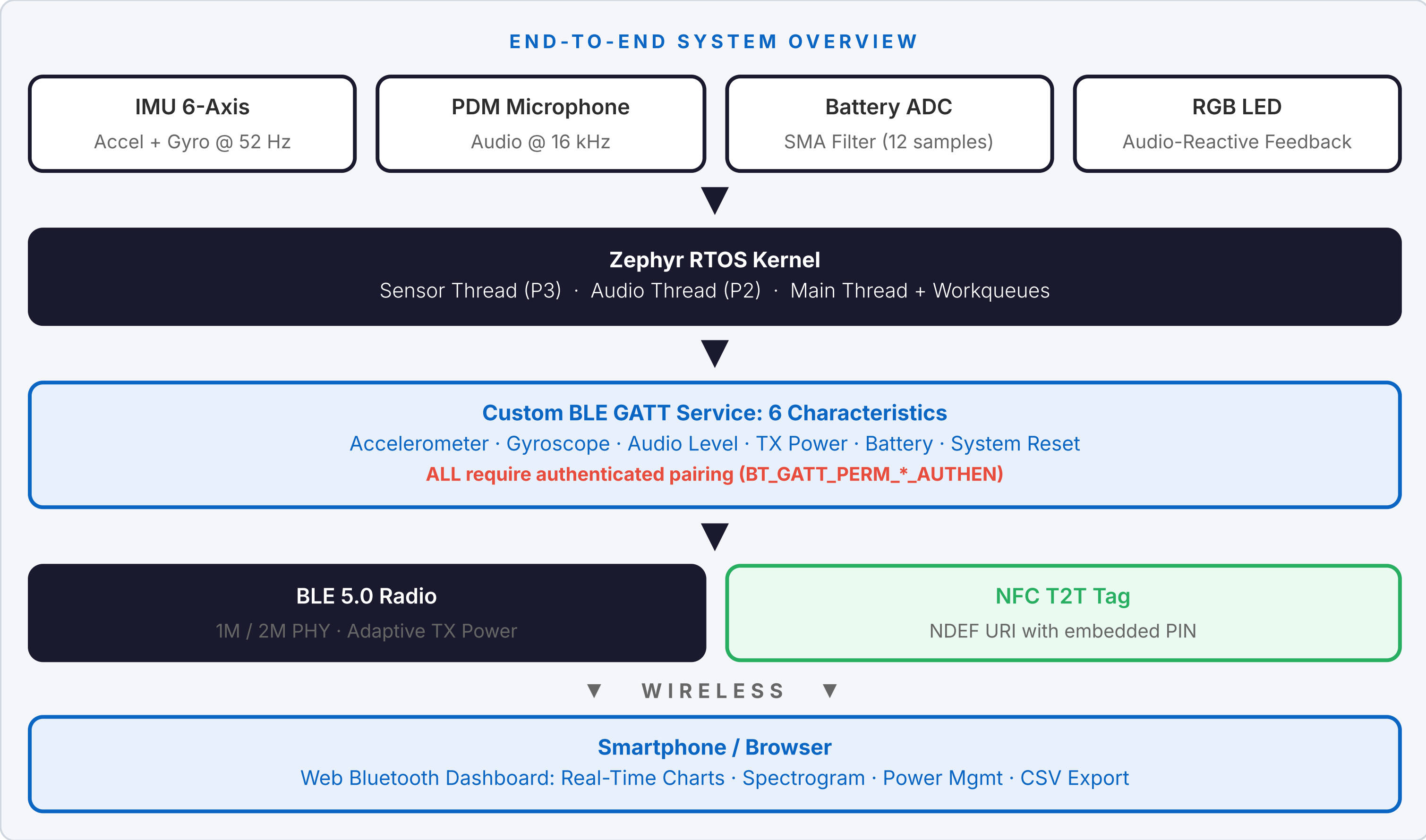


A **professional prototype sensor platform** built on the **Nordic nRF52840** (ARM Cortex-M4F) running **Zephyr RTOS**, demonstrating a fully headless multi-sensor BLE beacon architecture. The implementation addresses a key IoT UX challenge: **secure device onboarding without physical user interface**: by combining NFC Type 2 Tag emulation with hardware-derived cryptographic pairing, enabling seamless tap-to-connect operation from any NFC-enabled smartphone.

TECHNOLOGY STACK

LAYER	TECHNOLOGY
MCU / SoC	Nordic nRF52840: ARM Cortex-M4F, 1 MB Flash, 256 KB RAM
RTOS	Zephyr RTOS (preemptive multithreading, workqueue scheduling)
Connectivity	Bluetooth 5.0 LE (1M / 2M PHY), NFC Type 2 Tag (NDEF)
Security	BLE Security Level 4: AES-128 CCM, ECDH key exchange
Sensors	LSM6DS3TR-C 6-axis IMU (I^2C), PDM MEMS Microphone
Power	LiPo 3.3–4.15 V, BQ25100 charge IC, adaptive TX power
Interface	Web Bluetooth API dashboard: zero native app required
Hardware	Seeed Studio XIAO nRF52840 Sense

SYSTEM ARCHITECTURE



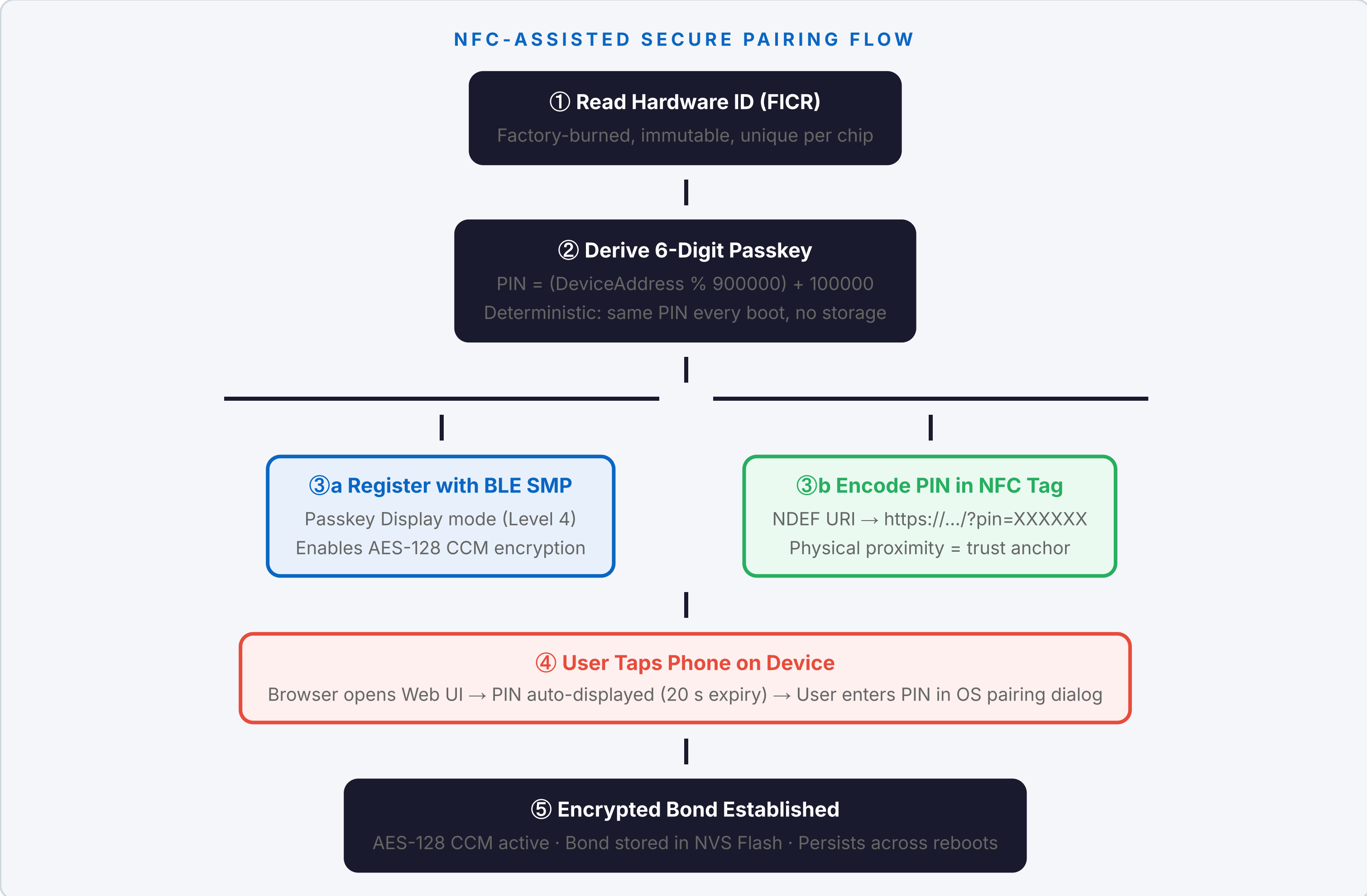
SECURITY ARCHITECTURE & NFC WORKFLOW

The Challenge: Secure Pairing Without a Screen

Traditional BLE devices display a pairing PIN on a screen or use insecure "Just Works" pairing. A headless sensor node has neither a display nor input: yet must enforce strong authentication to protect sensor data from unauthorized access.

The Solution: Hardware-Derived PIN via NFC Handoff

A three-layer security architecture turns NFC into a physical-proximity authentication channel:



GATT-Level Security Enforcement

Every characteristic enforces `READ_AUTHEN` / `WRITE_AUTHEN` permissions:

PROTECTION	EFFECT
No data leaks	Unauthenticated observers cannot read any sensor values
No unauthorized control	Remote bootloader reset requires a bonded, encrypted connection
Mandatory encryption	All traffic is AES-128 CCM encrypted after successful pairing
Persistent bonding	Credentials survive reboots via Zephyr NVS/Settings subsystem

Comparison: Traditional vs. This Approach

TRADITIONAL APPROACH	THIS SOLUTION
PIN printed on label: static, easily shared	PIN derived from silicon: unique per device
"Just Works" pairing: zero authentication	Passkey Display (Level 4): MITM-resistant
Requires companion app for onboarding	NFC tap opens standard browser: zero app install
PIN visible only during setup wizard	PIN in URL, auto-expires in Web UI (20 seconds)

REAL-TIME SYSTEM ARCHITECTURE

Preemptive Multi-Threading Model

Three concurrent threads on Zephyr's preemptive scheduler plus two asynchronous workqueue tasks for non-blocking power management:

ZEPHYR RTOS THREADING & DATA PIPELINE

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Audio Thread
PDM Microphone → 512-sample blocks @ 16 kHz → **RMS** (loudness) + **ZCR** (zero-crossing rate) → BLE GATT Notify → RGB LED audio-reactive color mapping (silence → blue → green → yellow → red)

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Sensor Thread
LSM6DS3TR-C IMU → Accel (X/Y/Z) + Gyro (X/Y/Z) @ 52 Hz → Little-endian int32 serialization → BLE GATT Notify to all connected clients

MAIN

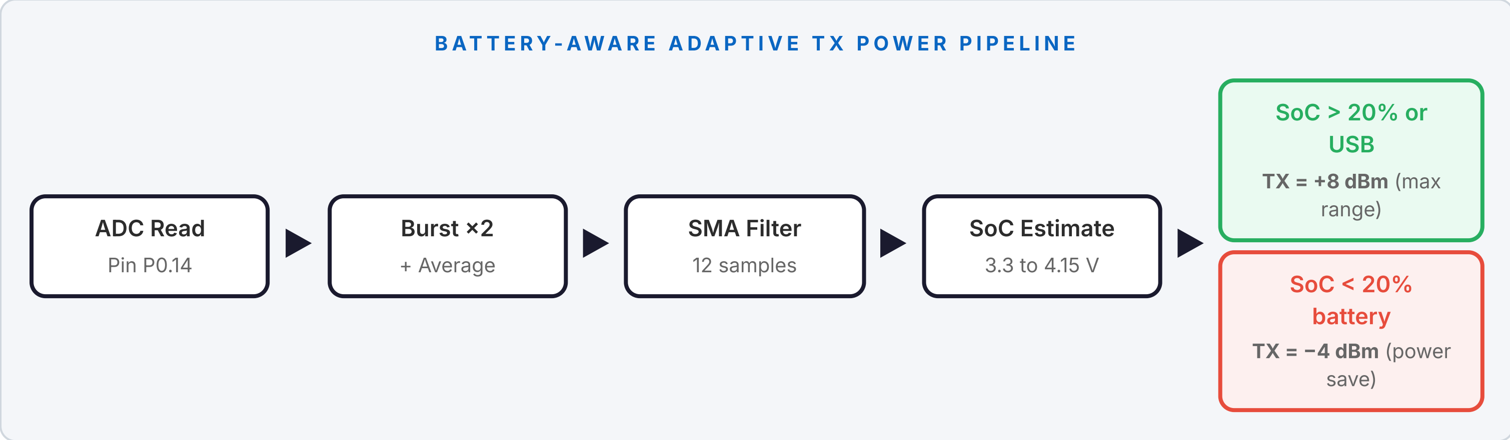
Main Thread + Delayed Workqueues
Battery Work: ADC burst → SMA(12) filter → SoC% estimation · Interval: 1 s (boot) → 10 s (steady)
Status Work: BQ25100 charge IC monitoring · USB VBUS detection · Charge current control (1 s poll)
TX Power Logic: SoC > 20% or USB → +8 dBm (max range) · SoC < 20% battery → −4 dBm (power save)

Custom BLE GATT Service

One primary service, six characteristics covering the full sensor pipeline and system control:

CHARACTERISTIC	UUID SUFFIX	TYPE	PAYLOAD
Accelerometer	...def1	Notify	3× int32 LE: X, Y, Z in milli-m/s^2
Gyroscope	...def2	Notify	3× int32 LE: X, Y, Z in milli-°/s
Audio Level	...def3	Notify	uint32 RMS + uint32 ZCR per block
TX Power	...def4	Read / Notify	int8: current power in dBm
Battery	...def5	Read / Notify	uint16 mV + uint8 SoC% + uint8 status
System Reset	...def6	Write	Write 0x01 → reboot to UF2 bootloader

Intelligent Power Management



WEB BLUETOOTH DASHBOARD

A fully responsive browser-native monitoring interface via the **Web Bluetooth API**: zero native app installation required. Works on Chrome and Edge across all platforms.

FEATURE	TECHNICAL DETAIL
Real-Time Telemetry	Acceleration & gyroscope rolling charts (Chart.js, 150-point buffer, pinch-to-zoom)
Audio Analysis	1024-point FFT, A-weighted Power Spectral Density, live Canvas 2D waterfall spectrogram
Power Management	Battery voltage, SoC %, charge status indicator, TX power with dBm → mW conversion
NFC PIN Integration	URL parameter `?pin=XXXXXX` auto-displays passkey with 20 s animated expiry timer
Remote System Control	One-click reboot to UF2 bootloader for wireless firmware updates
Performance	Visibility API throttling, 30 FPS render cap, EMA-smoothed audio, tab-suspend protection
Export	CSV telemetry session export for offline analysis

KEY ENGINEERING HIGHLIGHTS

1. **Zero-UI Onboarding**
NFC tag + Web Bluetooth = no display, no companion app, no manual PIN entry. Tap the device and go.

2. **Hardware-Rooted Trust**
Passkey derived from immutable FICR silicon ID. Deterministic: no RNG, no provisioning step, no key storage.

3. **Defense in Depth**
NFC (physical proximity) → BLE SMP (cryptographic gate) → GATT AUTHEN (per-characteristic access control).

4. **Battery-Aware Radio**
Dynamic TX power preserves energy at low SoC without sacrificing range when power is available.

5. **Non-Blocking Architecture**
All I/O is threaded or workqueue-driven. Main loop sleeps at 1 Hz. Zero busy-wait patterns anywhere.

6. **Bluetooth 5.0 Optimized**
2M PHY, extended data length (69 B), optimized intervals (15–30 ms), up to 3 simultaneous connections.

7. **Production Patterns**
NVS-backed bonding, SMA-filtered ADC, graceful BLE reconnect, RGB startup self-test sequence.

8. **Remote Firmware Update**
Wireless bootloader reset via authenticated BLE write: no physical access to the device required.

Zephyr RTOS · Nordic nRF52840 · BLE 5.0 · NFC T2T · Web Bluetooth · AES-128 CCM · Cortex-M4F · NDEF